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DEPARTMENT OF DEFENSE INTERFACE STANDARD

COOLING WATER FOR SUPPORT OF ELECTRONIC EQUIPMENT



AMSC N/A FSC 5975

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FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Purpose. This section defines the standard interface requirements for, and the constraints on, the design of shipboard electronic equipment that will utilize the ship's electronic cooling water system.

3. Nature of the interface. Navy ships have a requirement for cooling water to support electronic equipment. This is provided by a system or multiple systems specifically designed to supply the necessary quantities of conditioned water with standard characteristics. Cooling water is generated and then distributed to satisfy the needs of the various users. This section, MIL-STD-1399-532, provides guidance to ensure compatibility between each user and the cooling water system involved and for the determination of user requirements to assist in the design of such cooling water systems.

4. Structure. The technical content first describes the characteristics of the shipboard electronic cooling water system in terms of purity, temperature, pressure, flow rate, and thermal design. Constraints on user systems and the requirements for determination of user needs are then established.

5. Numerical quantities. Numerical quantities stated in this section are expressed in metric (SI) units and may be followed by inch-pound units in parentheses.

6. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard D.C. 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This section establishes interface requirements for shipboard electronic equipment utilizing an electronic cooling water system to ensure compatibility between such equipment and the electronic cooling water system. This section is applicable to electronic cooling water systems cooled by either seawater or by the ship's chilled water system. The policies and procedures established by MIL-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document. This interface Standard does not apply to electronic cooling systems under the technical cognizance of Naval Nuclear Propulsion (NAVSEA 08), unless specifically invoked.

1.2 Interface. Basic characteristics and constraint categories concerned with this interface are shown symbolically on figure 1 (see "Definitions" section of MIL-STD-1399). The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

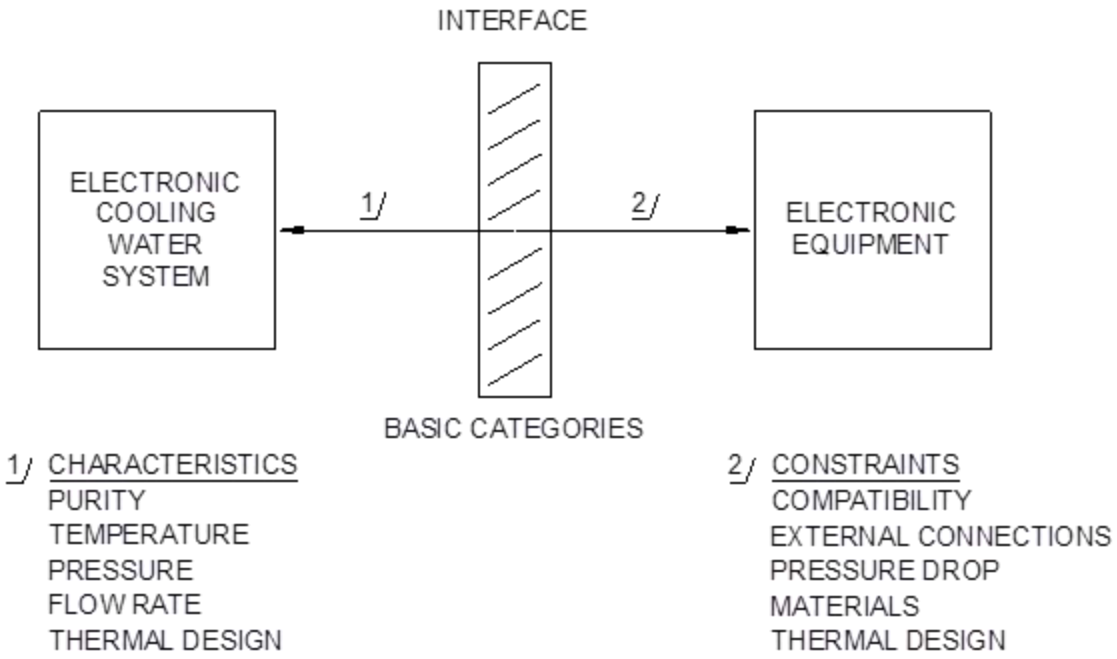


FIGURE 1. Interface.

1.3 Applicability. The criteria of this section apply when electronic equipment installed on ships or submarines requires water cooling in order to properly dispose of generated heat.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-T-24107	-	Tube, Copper (Seamless) (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200, and C14200)
MIL-C-24679	-	Copper-Nickel Alloy Forgings and Forging Stock
MIL-P-24691/3	-	Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1399	-	Interface Standard for Shipboard Systems

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S6430-AE-TED-010	-	Piping Devices, Flexible Hose Assemblies, Volume 1; Technical Directive

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A182/A182M	-	Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A312/A312M	-	Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A403/A403M	-	Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings
ASTM B505/B505M	-	Standard Specification for Copper Alloy Continuous Castings

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ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Conduction cooling. The transfer of heat from one hot object to another cooler object by direct contact. In electronic cooling, this is accomplished by water or other cooling media flow in chassis plates or mounting panels.

3.2 Convection cooling. A mechanism where heat is transferred from a hot object by the flow of a fluid (liquid or gas) surrounding the object. In electronic cooling, this is accomplished by use of a circulating forced air system consisting of heat dissipating fins, air channels, blowers, and air-to-water heat exchangers.

3.3 Deionized water. Synonym of demineralized water.

3.4 Demineralization. Demineralization of water is the removal of all inorganic salts by ion exchange. In this process, a strong acid cation resin converts dissolved salts into their corresponding acids, and a strong base anion resin removes these acids.

3.5 Demineralized water. Water having less than 10 parts per million (ppm) total dissolved solids (TDS), a chloride concentration no greater than 0.065 equivalents per million (elm), maximum conductivity of 2 microsiemens, an oxygen concentration no greater than 0.5 ppm by weight, and particulates no greater than 0.5 micrometers in size. This purity is achieved by demineralizing.

3.6 Distilled water. Distilled water plants create a phase change in the source feed water, changing a portion of it to steam, and concentrating the liquid into brine containing the salts and other material that had been present in the water. The steam is collected and condensed. This condensate is distilled water.

3.7 Fresh water. Fresh water is desalinated water or natural water capable of being made drinkable. Fresh water is not potable unless it is safe for human consumption. Fresh water is generally characterized by having low concentrations of dissolved salts and other total dissolved solids.

3.8 High purity water. Synonym of demineralized water.

3.9 Shipboard electronic cooling water system. A system specifically designed to remove heat from electronic equipment by supplying necessary quantities of conditioned water with standard characteristics as specified herein.

3.10 User equipment or system. Any electronic system or equipment that utilizes the shipboard electronic cooling water system.

4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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4.2 Requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to by Systems Commands (SYSCOMs), project managers, contractors, and all others engaged in any aspect of shipboard electronic cooling water system design to which these requirements and constraints apply, including equipment design, production, and installation (see "General Requirements" section of MIL-STD-1399).

5. DETAILED REQUIREMENTS

5.1 General considerations. Shipboard electronic equipment generates heat during normal operations. This heat must be removed to prevent damage to heat-critical components. Heat removal may be by natural dissipation, air cooling, or where these measures are not adequate to remove the required amount of heat, by an electronic cooling water system. Forced air cooling is the preferred cooling method. When the inlet and outlet temperature differential exceeds 14 degrees Celsius (°C) (57 degrees Fahrenheit [°F]), water cooling shall be used. Water cooling systems may employ conduction cooling, convective cooling, or other means, such as direct immersion and vaporization cooling. Convection is the preferred method as long as all cooling requirements can be met. This section concerns the cooling water systems employing cool water and conduction or convection cooling (or a combination of those) composed of the necessary pumps, heat exchangers, demineralizers, water flow controls, pressure relief valves, temperature regulating valves, temperature, flow and pressure indicators, conductivity indicators, alarms, air trap and expansion tank, stop valves, piping, and all other accessories required to ensure a continuous flow of cooling water of specified characteristics to the electronic equipment. Cooling of water for the electronic cooling water system is via heat exchangers using water from the sea water system or the chilled water system. Electronic cooling systems employing direct immersion, vaporization cooling, or other methods shall be approved by NAVSEA prior to use. Such systems are not covered by this standard.

5.2 Interface characteristics. The interface characteristics of the ship's cooling water system shall be in accordance with 5.2.1 through 5.2.5. Each shipboard system shall deliver cooling water with these characteristics to the electronic equipment interface. High purity reverse osmosis (HPRO) water, 0.065 elm chlorides maximum, will normally be used for filling electronic cooling water systems. In certain installations, demineralized water from off-ship sources may be required; such water supplied to the electronic cooling systems also shall meet the requirements of 5.2.1 through 5.2.5.

5.2.1 Purity. Unless otherwise specified (see 6.2), the purity characteristics shall be as follows:

a. Conductivity. Not in excess of 2.0 microsiemens per centimeter (µS/cm) (micromhos per centimeter) corrected to 25 °C (77 °F). Some HPRO units can produce water of ≤l µS/cm conductivity.

b. Oxygen concentration. 0.5 ppm by weight maximum.

c. Filtration. Size of solids shall not exceed 0.5 micrometer. If required by the served electronic equipment, full-flow filtration and full deionization shall be provided.

5.2.2 Temperature. The temperature of the cooling water provided to the electronic equipment inlet connections shall be not greater than 40 °C (104 °F) on surface ships and not greater than 35 °C (95 °F) on submarines. The cooling water system can easily provide water at higher temperatures if required in special cases; when operating in cold sea water, lesser temperatures may be available. The ability of electronic equipment to use water at a temperature not greater than 40 °C (104 °F) on surface ships and 35 °C (95 °F) on submarines is extremely critical because of the added cost involved in supporting any requirement less than these temperatures. Such requirement involves the use of chilled water from the air conditioning system in lieu of the use of sea water for the heat exchanger. This would require significant increases in cost, weight, space, and a potential compromise in reliability of the air conditioning system. Certain systems may require different heat and flow ranges specific to their requirements. Any need to deviate from this temperature interface characteristic shall be fully justified (see 6.2).

5.2.3 Pressure. For surface ships, the water pressure shall be within the range of 69 kilopascals (kPa) (10 pounds per square inch [lb/in^2]) to 758 kPa (110 lb/in^2). The equipment cooling system shall be capable of withstanding a hydrostatic pressure test of 1,034 kPa (150 lb/in^2). For submarine service, the water pressure shall be within the range of 69 kPa (10 lb/in^2) to 1,415 kPa (205 lb/in^2). The equipment cooling system shall be capable of withstanding a hydrostatic pressure test of 1,910 kPa (277 lb/in^2).

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5.2.4 Flow rate. The flow rate shall be as required by the electronic equipment and controlled by the design of the equipment water flow rate and distribution. The design shall be such as to maximize laminar flow conditions and to minimize turbulent flow.

a. Water flow rate. Unless otherwise specified (see 6.2), the total water flow rate from the shipboard cooling water system shall be adjusted to be a maximum of 5.3 liters per minute (L/min) (1.4 gallons per minute [gal/min]) per kilowatt load based on a maximum of 2 °C (3.6 °F) temperature rise in the water.

b. Water distribution. The distribution of water from the shipboard cooling water system shall be apportioned in accordance with the heat dissipation of each heat exchanger, chassis, or panel served. The water distribution shall be controlled by the cross-sectional area of the water passages as much as possible. Any additional adjustments necessary shall be achieved with fixed built-in flow devices in the connecting tubing.

5.2.5 Thermal design. The thermal design of an electronic cooling water system shall provide heat flow paths from heat sources to the cooling system that are as short, direct, and as low a thermal resistance as practicable. Such means shall be consistent with the performance, operating, and maintenance requirements specified in the individual equipment specification and as determined in 5.3.5. Construction and placement of parts shall be such that minimum chance of leakage exists and that, should a leak occur, there is a minimum possibility of water striking electronic parts or electrical circuitry. The operating temperature of heat-dissipating and heat-sensitive parts shall be within their thermal stress range and shall be derated, if necessary. The thermal design shall be robust enough to ensure that the cooled equipment can operate at full power within its temperature parameters for at least 8 hours at an ambient temperature of 35 °C (95 °F) without any new supply of cooling water.

5.3 Interface constraints. The interface characteristics of the shipboard electronic cooling water system impose certain constraints on the design of electronic equipment utilizing such support. These constraints are specified in 5.3.1 through 5.3.6.

5.3.1 Compatibility. The design of electronic equipment utilizing a ship's cooling water support shall be compatible with the interface characteristics given in 5.2.

5.3.2 Pressure drop. The maximum pressure drop from inlet to outlet of each electronic cabinet shall not exceed 69 kPa (10 lb/in^2). While a high-pressure piping system required to support large pressure drops is not desirable from the point of view of the additional weight and cost involved, it may be a cost-effective compromise that warrants a deviation from this interface constraint. In such instance, the need to deviate from this pressure drop interface shall be fully justified (see 6.2).

5.3.3 Coolant circuit materials. Unless otherwise specified (see 6.2), the coolant circuit shall be constructed of the materials identified in table I and electrochemically compatible with other materials in the system. Materials containing carbon are not acceptable. Flexible hose materials shall be selected in accordance with guidance provided in S6430-AE-TED-010.

TABLE I. Coolant circuit materials.

Item	Material	Applicable Documents
Pipe/tube	Copper-nickel C70600 or C71500	MIL-T-16420
	Copper	MIL-T-24107
	Bronze C92200 or C90300	ASTM B505/B505M, ASTM B584, or ASTM B61
	Corrosion-resistant steel (CRES) 304, 304L, or 316L	MIL-P-24691/3 or ASTM A312/A312M

NOTES:
1. Surface ship systems using distilled or fresh water as coolant shall use copper or copper alloy materials.
2. Copper piping shall not be used in water systems where copper carbonate fouling would be a problem, such as in demineralized water systems.

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5.3.4 External connections. Unless otherwise specified (see 6.2), each unit shall have a single supply and single return water connection for attachment to the electronic cooling water system, and connections shall be rigidly fastened to the outside surface of the enclosure.

5.3.4.1 Rigidly-mounted equipment. Rigidly-mounted equipment shall use flanged or union connections in accordance with table II for surface ships and table III for submarines.

5.3.4.2 Shock- and resiliently-mounted equipment. Shock- or resiliently-mounted equipment shall be provided with either flexible hose assemblies or flanged or union hose connections in accordance with table II for surface ships and table III for submarines (see 6.2). Selection of hoses and hose-threaded connections (if employed) shall be in accordance with S6430-AE-TED-010.

TABLE II. Surface ship external connections.

Item	Material	Applicable Documents
Fittings – unions	Bronze C92200 or C90300	ASTM B505/B505M, ASTM B584, or ASTM B61
	CRES 304L	ASTM A182/A182M
Flanges	Bronze C92200 or C90300	ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61
	Copper-nickel C70600 or C71500	MIL-C-15726 or MIL-C-24679
	CRES 304L	ASTM A182/A182M

TABLE III. Submarine external connections.

Item	Material	Applicable Documents
Fittings – unions	Copper-nickel C71500	MIL-C-15726 or MIL-C-24679
	Bronze C92200 or C90300	ASTM B271/B271M, ASTM B505/B505M, ASTM B584, or ASTM B61
	CRES 304	ASTM A403/A403M
Flanges	Bronze C92200 or C90300	ASTM B505/B505M, ASTM B584, or ASTM B61
	CRES	ASTM A182/A182M, ASTM A403/A403M, or SAE AMS-QQ-S-763

5.3.5 User cooling water requirements. The cooling water requirements of each individual user system or equipment and those of the entire ship or submarine shall be determined (see 6.2) and utilized in the thermal design within the interface requirements of 5.2.1 through 5.2.5 and 5.3.2. These requirements are as follows:

a. Cooling water requirements – individual user system or equipment. The following parameters shall be determined for each user equipment or system that will utilize cooling water from a shipboard electronic cooling water system:

(1) Nominal operating pressure – kPa (lb/in^2).

(2) Maximum operating pressure – kPa (lb/in^2).

(3) System design pressure – kPa (lb/in^2).

(4) Flow rate – L/min (gal/min) (average per unit).

(5) Flow rate – L/min (gal/min) (maximum per unit).

(6) Purity, if other than specified herein.

(7) Supply temperature, if other than specified herein (see 5.2.2).

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(8) Pressure drop – kPa (lb/in^2).

b. Cooling water requirements – total ship. The following parameters shall be determined for the total ship electronic cooling water support system:

(1) User equipment or systems.

(2) Number of user's equipment or systems installed.

(3) Nominal operating pressure – kPa (lb/in^2).

(4) Maximum operating pressure – kPa (lb/in^2).

(5) System design pressure – kPa (lb/in^2).

(6) Total flow rate – L/min (gal/min).

(7) Use factor (percent of total ship electronic cooling load apportioned to each equipment).

(8) Purity, if other than specified herein.

(9) Temperature, if other than specified herein.

5.3.6 Thermal design. Thermal design constraints shall be imposed by thermal sensors at appropriate locations inside the equipment to detect abnormal temperatures and to operate automatic alarms and protective devices. The number and location of these devices shall be such that failure of any portion of the heat removal system shall be detected. The temperature-sensing device shall render the configuration item inoperative but shall not disable any internal cooling devices. Thermal sensor constraints shall be capable of being overridden by operators for emergency conditions.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The water cooling systems covered by this standard are intended for cooling of electronic equipment onboard naval ships and submarines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Water purity requirements, if not as specified herein (see 5.2.1).

c. Justification for deviation from temperature interface characteristic (see 5.2.2).

d. Maximum water flow rate, if not as specified herein (see 5.2.4.a).

e. Justification for deviation from pressure drop interface (see 5.3.2).

f. Materials, if not as specified herein (see 5.3.3 and 5.3.4).

g. Identification of connections by hard pipe (or by hose for shock) or resiliently-mounted equipment (see 5.3.4).

h. Determination of cooling water requirements for each user system and equipment, and consideration of growth margin over the anticipated lifespan of those (see 5.3.5).

6.3 Subject term (key word) listing.

Conduction

Convection

Immersion

Thermal

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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